

11. Build & Deployment

11.1 Overview

Build and Deployment is the process of preparing the automation framework for execution and deploying the latest application build into the test environment for validation.

In this project, Maven is used for build management, while Selenium WebDriver and TestNG are used for automated test execution.

11.2 Build Process

Step 1: Source Code Development

The automation scripts are developed using:

- Java
- Selenium WebDriver
- TestNG
- Maven

The project follows the Page Object Model (POM) design pattern.

Step 2: Code Compilation

Maven compiles all Java source files present in:

```
src/main/java  
src/test/java
```

Command:

```
mvn clean compile
```

Expected Result:

- All Java files compile successfully.
 - No compilation errors occur.
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Step 3: Dependency Management

Maven automatically downloads required dependencies from the Maven Repository.

Dependencies used:

- Selenium Java
- TestNG
- WebDriverManager

Benefits:

- Automatic library management
 - Version control
 - Reduced manual effort
-

Step 4: Test Execution Build

Before execution, Maven creates a test build.

Command:

```
mvn test
```

Activities Performed:

- Browser initialization
 - TestNG execution
 - Result generation
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11.3 Deployment Environment

Application Under Test

Parameter	Value
Application	OrangeHRM Demo
Type	Web Application
Environment	Test Environment
URL	https://opensource-demo.orangehrmlive.com

11.4 Test Deployment Process

Deployment Steps

1. Launch Browser
2. Open OrangeHRM Application URL

3. Verify Application Availability
4. Execute Automation Scripts
5. Generate Test Results

11.5 Build Validation Testing (BVT)

After deployment, Build Validation Testing is performed.

Objectives

Verify:

- Application launches successfully
- Login page loads correctly
- Dashboard loads correctly
- Navigation works properly
- Forgot Password page is accessible

11.6 Continuous Testing Benefits

Although this project is executed manually through Eclipse/TestNG, it can be integrated with:

- Jenkins
- GitHub Actions
- Azure DevOps

Benefits:

- Automated execution
- Faster feedback
- Continuous Integration support

11.7 Build Status

Build ID Version Status

Build_01 v1.0 Passed

11.8 Build Deployment Outcome

The application build was successfully deployed to the test environment and all automation test scripts executed successfully without major issues.
